

DM545/DM871 – Linear and integer programming

Exercise Sheet 5, Autumn 2026

Exercises with the symbol $+$ are to be done at home before the class. Exercises with the symbol $*$ will be tackled in class and should be at least read at home. The remaining exercises are left for self training after the exercise class.

Exercise 1 $^+$

Using the Lagrangian multipliers method we showed in class that

$$\text{the dual of } \min\{c^T x \mid Ax = b, x \geq 0\} \text{ is } \max\{b^T y \mid A^T y \leq c\}.$$

Using the same method, show that the dual of

$$\text{the dual of } \max\{c^T x \mid Ax \leq b, x \geq 0\} \text{ is } \min\{b^T y \mid A^T y \geq c, y \geq 0\}.$$

Exercise 2 $*$

Consider the following problem:

$$\begin{array}{ll} \text{maximize} & z = x_1 - x_2 \\ \text{subject to} & x_1 + x_2 \leq 2 \\ & 2x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{array}$$

In the ordinary simplex method this problem does not have an initial feasible basis. Hence, the method has to be enhanced by a preliminary phase to attain a feasible basis. Traditionally we talk about a *phase I–phase II* simplex method. In phase I an initial feasible solution is sought and in phase II the ordinary simplex is started from the initial feasible solution found.

There are two ways to carry out phase I.

- Solving an auxiliary LP problem defined by introducing auxiliary variables and minimizing them in the objective. The solution of the auxiliary LP problem gives an initial feasible basis or a proof of infeasibility.
- Applying the dual simplex on a possibly modified problem to find a feasible solution. If the initial infeasible tableau of the original problem is not optimal then the objective function can be temporarily modified for this phase in order to make the initial tableau optimal although not feasible. Opposite to the primal simplex method, the dual simplex method iterates through infeasible basic solutions, while maintaining them optimal, and stops when a feasible solution is reached.

Dual Simplex: The strong duality theorem states that we can solve the primal problem by solving its dual. You can verify that applying the *primal simplex method* to the dual problem corresponds to the following method, called *dual simplex method* that works on the primal problem:

1. (Feasibility condition) select the leaving variable by picking the basic variable whose right-hand side term is negative, i.e., select i^* with $b_{i^*} < 0$.
2. (Optimality condition) pick the entering variable by scanning across the selected row and comparing ratios of the coefficients in this row to the corresponding coefficients in the objective row, looking for the largest negated. Formally, select j^* such that $j^* = \min\{|c_j/a_{i^*j}| : a_{i^*j} < 0\}$
3. Update the tableau around the pivot in the same way as with the primal simplex.

4. Stop if no right-hand side term is negative.

Duality can help us with the issue of initial feasible basic solutions. In the problem above, the initial tableau is infeasible and not optimal, hence we cannot apply the primal simplex nor the dual simplex. However, if the objective function was $w = -x_1 - x_2$, then we would have the conditions of infeasibility and optimality needed by the dual simplex. You can understand this also looking at the dual problem. With the new objective function the initial basic solution of the dual problem would be feasible and we could solve the problem solving the dual problem with the primal simplex. In contrast, with original objective function z the primal simplex has infeasible initial basis in both problems. So we can change temporarily the objective function z with w and apply the dual simplex method to the primal problem. When it stops we reached a feasible solution that is optimal with respect to w . We can then reintroduce the original objective function and continue iterating with the primal simplex. The phase I–phase II simplex method that uses the dual simplex is also called the *dual-primal simplex method*.

Apply the two versions of the phase I–phase II simplex method (that is, phase I is carried out with the auxiliary problem or with the dual simplex) to the problem above and verify that they lead to the same solutions.

Exercise 3⁺

Solve the systems $\mathbf{y}^T E_1 E_2 E_3 E_4 = [1 \ 2 \ 3]$ and $E_1 E_2 E_3 E_4 \mathbf{d} = [1 \ 2 \ 3]^T$ with

$$E_1 = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0.5 & 0 \\ 0 & 4 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \quad E_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad E_4 = \begin{bmatrix} -0.5 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Exercise 4* Sensitivity Analysis and Revised Simplex

A furniture-manufacturing company can produce four types of products using three resources.

- A bookcase requires three hours of work, one unit of metal, and four units of wood and it brings in a net profit of 19 Euro.
- A desk requires two hours of work, one unit of metal and three units of wood, and it brings in a net profit of 13 Euro.
- A chair requires one hour of work, one unit of metal and three units of wood and it brings in a net profit of 12 Euro.
- A bedframe requires two hours of work, one unit of metal, and four units of wood and it brings in a net profit of 17 Euro.
- Only 225 hours of labor, 117 units of metal and 420 units of wood are available per day.

In order to decide how much to make of each product so as to maximize the total profit, the managers solve an LP problem.

- 1) Write the mathematical programming formulation of the problem.
- 2) With the help of a computational environment such as Python for carrying out linear algebra operations, write the optimal tableau, which has x_1, x_3 and x_4 in basis.

Do this task:

- first, using the original simplex method
 - second, using the revised simplex. In this case, start by writing A_B, A_N , then calculate $A_B^{-1}A_N$, and the finally derive the optimal tableau and verify that the solution is indeed optimal.
- 3) What is the increase in price (reduced cost) that would make product x_2 worth to be produced?
 - 4) What is the marginal value (shadow price) of an extra hour of work or amount of metal and wood?
 - 5) Are all resources totally utilized, i.e. are all constraints “binding”, or is there slack capacity in some of them? Answer this question in the light of the complementary slackness theorem.

- 6) From the economical interpretation of the dual why product x_2 is not worth producing? What is its imputed cost?

Perform a sensitivity analysis for the following variants:

- 7) The net profit brought in by each desk increases from 13 Euro to 15 Euro.
- 8) The availability of metal increases from 117 to 125 units per day
- 9) The company may also produce coffee tables, each of which requires three hours of work, one unit of metal, two units of wood and bring in a net profit of 14 Euro.
- 10) The number of chairs produced must be at most five times the numbers of desks

Exercise 5* Traffic Light Control

Automobile traffic from three highways, H1, H2, and H3, must stop and wait for a green light before exiting to a toll road. The tolls are 40 kr, 50 kr, and 60 kr for cars exiting from H1, H2, and H3, respectively. The flow rates from H1, H2, and H3 are 550, 650, and 450 cars per hour. The traffic light cycle may not exceed 2.2 minutes, and the green light on any highway must be at least 22 seconds. The yellow light is on for 10 seconds. The toll gate can handle a maximum of 500 cars per hour. Assuming that no cars move on yellow, and that clearly when a highway has the green signal the other two have the red, determine the optimal green time interval for the three highways that will maximize toll gate revenue per traffic cycle.

Exercise 6 Factory Planning and Machine Maintenance

Tasks 1–3 of [Factory Planning and Maintenance Case](#).